```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

file_path = r'C:\sales_data.csv'
df = pd.read_csv(r"C:\Users\kyoga\Downloads\sales_data - Sheet1.csv")

print(df.head())

print(df.isnull().sum())

df['Sales'] = df['Sales'].fillna(df['Sales'].mean())
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)

print(df.describe())

product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()

print(product_summary)

plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()

df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)
sales_over_time = df.groupby('Date').agg({
    'Sales': 'sum'
}).reset_index()

plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.show()

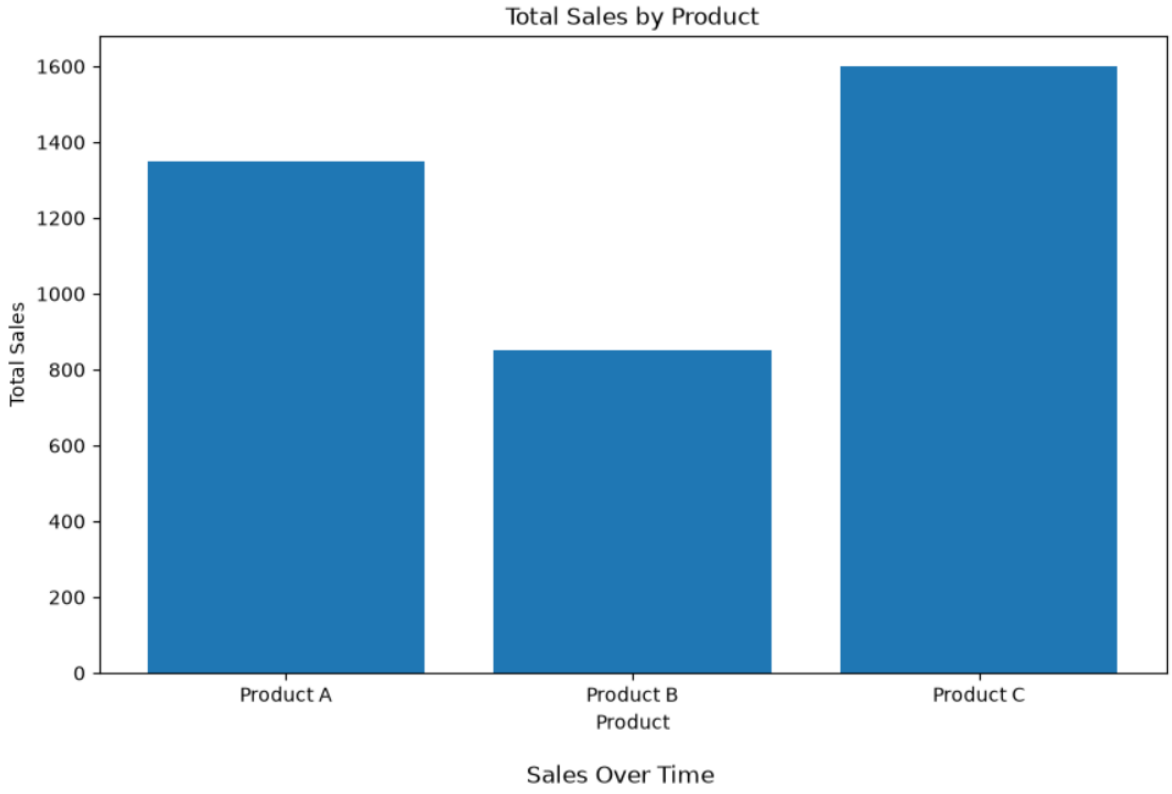
pivot_table = df.pivot_table(
    values='Sales',
    index='Region',
    columns='Product',
    aggfunc=np.sum,
    fill_value=0
)

print(pivot_table)

correlation_matrix = df.corr(numeric_only=True)
print(correlation_matrix)

plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

```
         Date     Product  Sales  Quantity Region
0  01-01-2023   Product A    200         4  North
1  02-01-2023   Product B    150         3  South
2  03-01-2023   Product A    220         5  North
3  04-01-2023   Product C    300         6   East
4  05-01-2023   Product B    180         4   West
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64
              Sales    Quantity
count    16.000000   16.000000
mean    237.500000    5.375000
std      64.031242    1.746425
min     150.000000    3.000000
25%     187.500000    4.000000
50%     225.000000    5.500000
75%     302.500000    7.000000
max     340.000000    8.000000
      Product  Sales  Quantity
0  Product A   1350        33
1  Product B    850        17
2  Product C   1600        36
```



Sales Over Time